

ECE 50877 INTRODUCTION TO APPLIED CRYPTOGRAPHY

EXAMPLE PROPOSALS

Example Project 1

This project proposal is based on Bernstein et al., “Sliding Right into Disaster: Left-to-right Sliding Windows Leak”, In International Conference on Cryptographic Hardware and Embedded Systems, 2017.

Title: Sliding Right into Disaster: Left-to-right Sliding Windows Leak

Problem Definition: Modular exponentiation is the computational core of RSA signing and decryption. Many cryptographic libraries (e.g., Libgcrypt, used by GnuPG) implement it using the left-to-right sliding-window method because it can be computed “on the fly,” without separate passes for recoding and exponentiation. Conventional wisdom holds that observing the sequence of squarings and multiplications through a cache side channel does not leak enough bits to enable a key-recovery attack. This estimate, however, is derived from the right-to-left windowing method. It remains an open question whether the left-to-right variant leaks meaningfully more information due to structural differences and whether that additional leakage is sufficient to mount a practical full private-key recovery attack against real-world implementations. This leads to the following research question: Does the direction of sliding-window recoding (left-to-right vs. right-to-left) materially change the amount of exponent information leaked through a square-and-multiply side channel, and if so, can this additional leakage be exploited to fully recover RSA private keys in practice?

Proposed Methodology: We will explore the information leakage under left-to-right recoding and analyze potential reasons that left-to-right may leak more bits than right-to-left. We extend the existing algorithm by Heninger and Shacham to make use of less readily available information to attack RSA-2048. We will validate the approach on large sets of randomly generated keys, then mount a real Flush+Reload cache-timing attack against Libgcrypt to recover live traces and attempt full private-key extraction.

Expected Findings: We expect left-to-right recoding to leak substantially more exponent information than right-to-left, enabling reliable full key recovery for RSA. We also expect the real-world side-channel attack to successfully recover complete traces from a small number of signatures, supporting a case for moving away from sliding-window exponentiation in constant-time-sensitive code.

Example Project 2

This project proposal is based on Chang et al., “HOLMES: Efficient Distribution Testing for Secure Collaborative Learning”, In USENIX Security Symposium, 2023.

Title: HOLMES: Efficient Distribution Testing for Secure Collaborative Learning

Problem Definition: Using secure multiparty computation (MPC), organizations that hold sensitive data (e.g., in healthcare, finance, or law enforcement) can jointly train machine learning models over their combined dataset without revealing their raw data to each other. However, secure computation restricts what operations can be performed on the joint dataset, which makes it difficult to assess the quality of that data before training on it. Without such an assessment, an organization cannot verify whether its collaborators’ data is skewed, biased, or otherwise unsuitable. At the same time, regulations such as the EU’s GDPR hold organizations legally responsible for errors or discrimination caused by their models. Distribution testing is a natural way to address this, but performing it under MPC has historically been prohibitively expensive. This leads to the following research question: Can distribution testing be performed efficiently enough under secure multiparty computation to be a practical, deployable step in real-world collaborative learning pipelines, and how much of a performance gain is achievable over existing MPC-only approaches?

Proposed Methodology: We will design a hybrid protocol, HOLMES, that combines MPC with interactive zero-knowledge (IZK) proofs so that per-party statistical checks are verified via lightweight pairwise IZK rather than expensive multi-party MPC. We will implement a library of standard statistical tests and benchmark it against generic MPC, pairwise 2PC, and NIZK/SNARK baselines on both simulated and real datasets.

Expected Findings: We expect HOLMES to substantially outperform all baselines, with at least an order-of-magnitude speedup for one-dimensional tests and several orders of magnitude for multidimensional tests. We also expect its accuracy on corrupted datasets to match that of standard non-private statistical tests, supporting its practicality as a routine step in secure collaborative learning.